



What Factors Drive Emergency Department Wait Times?

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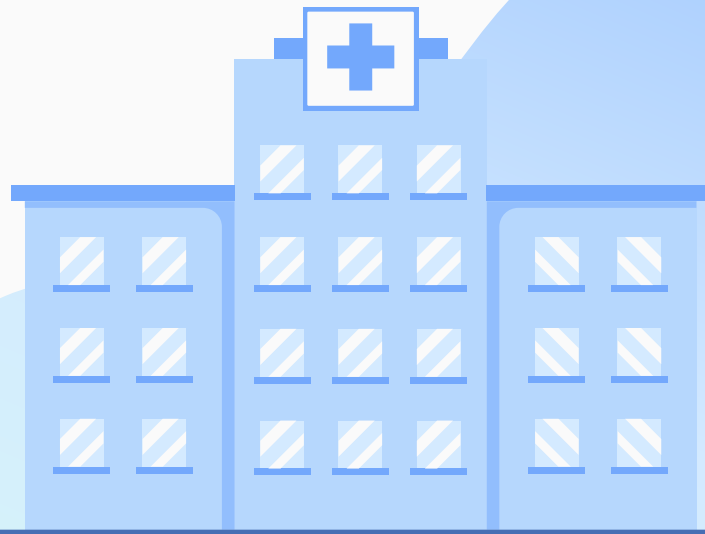


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Introduction



Why Understanding ED Wait Inequality Matters



1

James et al. (2005)
**U.S. pediatric emergency
departments**

- **Minority** children waited **longer than white children**, even after adjustment.
- Differences partly tied to lower-urgency triage assignments.
- **Shows inequities can occur within hospitals.**



2

Turner et al. (2022)
**England NHS (public,
free-at-point-use system)**

- Patients from deprived areas faced **consistent delays**.
- They **received less complex care** and **had higher re-attendance**.
- Suggests **roles for communication barriers or implicit bias**.



3

Swedish SES study
(2015–2019)
Chest Pain ED visits

- 140k+ patients; SES measured via income quintiles.
- Lowest-income group waited **~6 min longer** to doctor and **~11 min longer** LOS.
- **SES impacts wait time** even after clinical/triage adjustment.

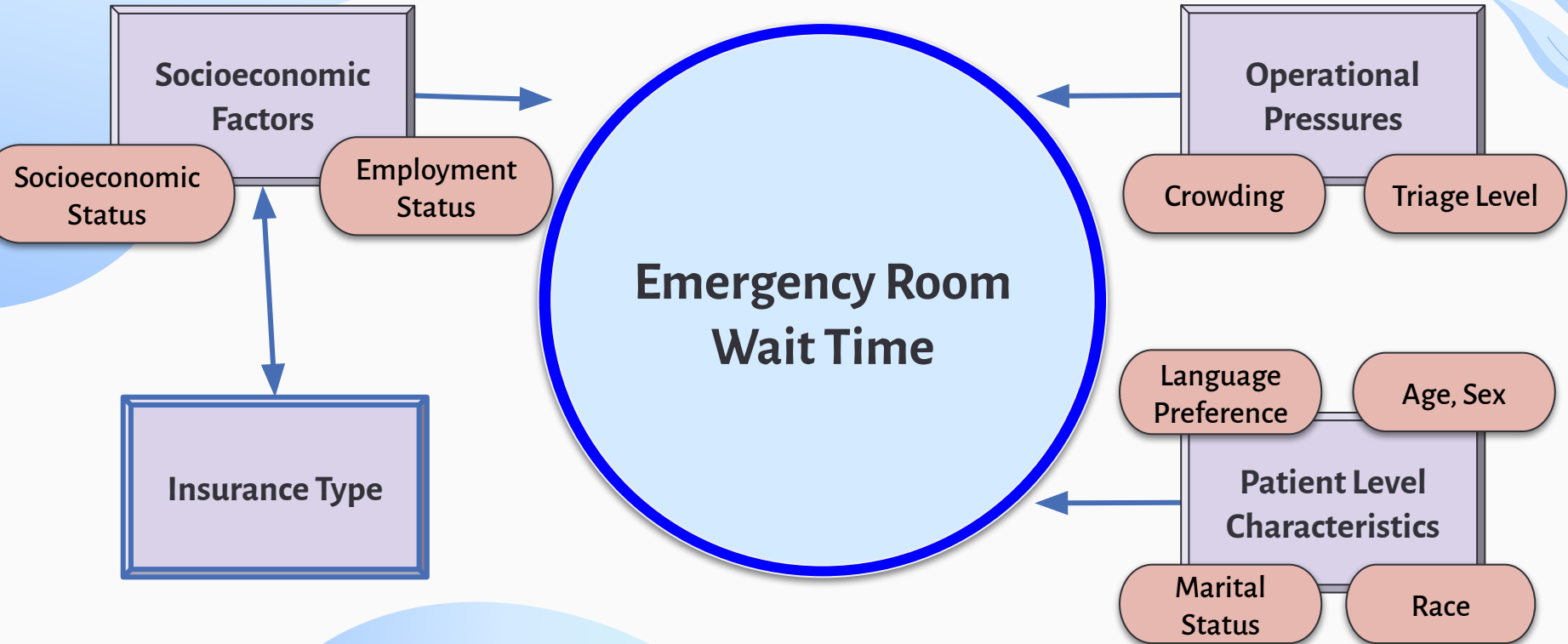
ED wait time reflects both operational crowding and potential inequities—motivating our analysis of predictors and interaction effects (e.g., SES × insurance)



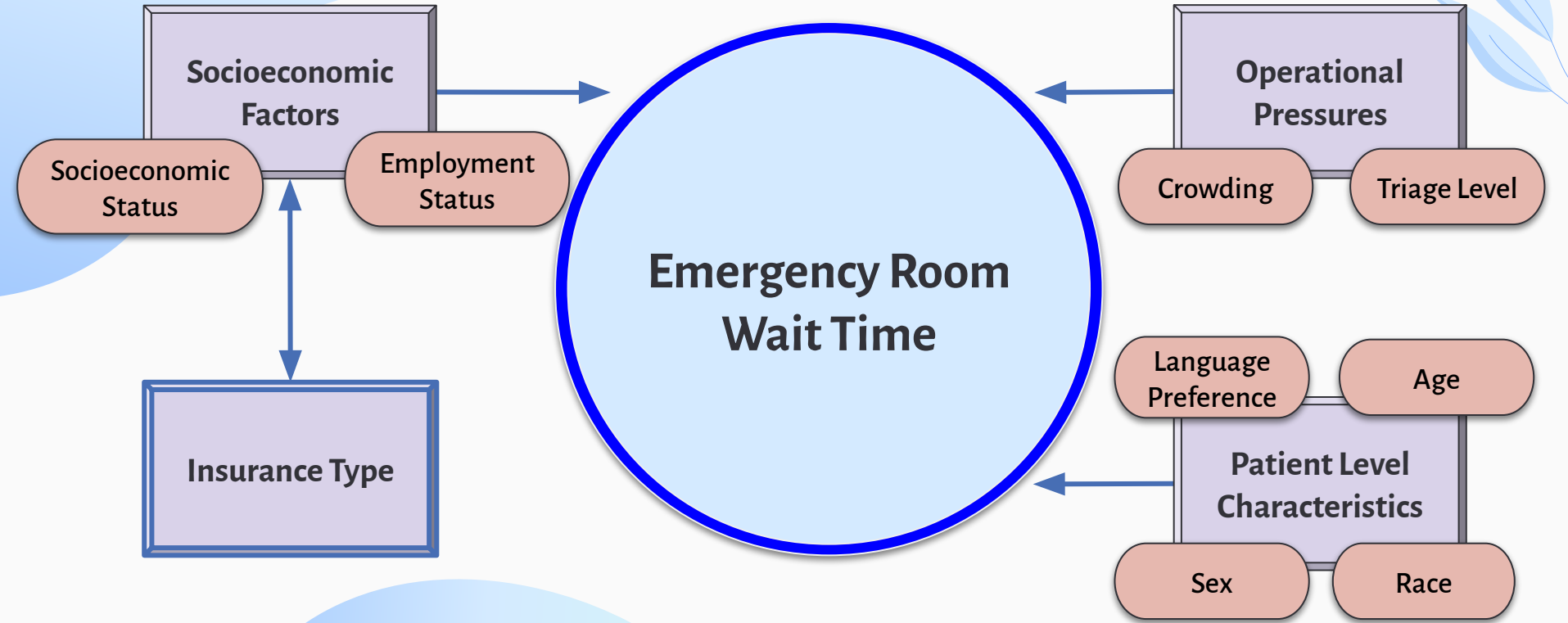
Research Question

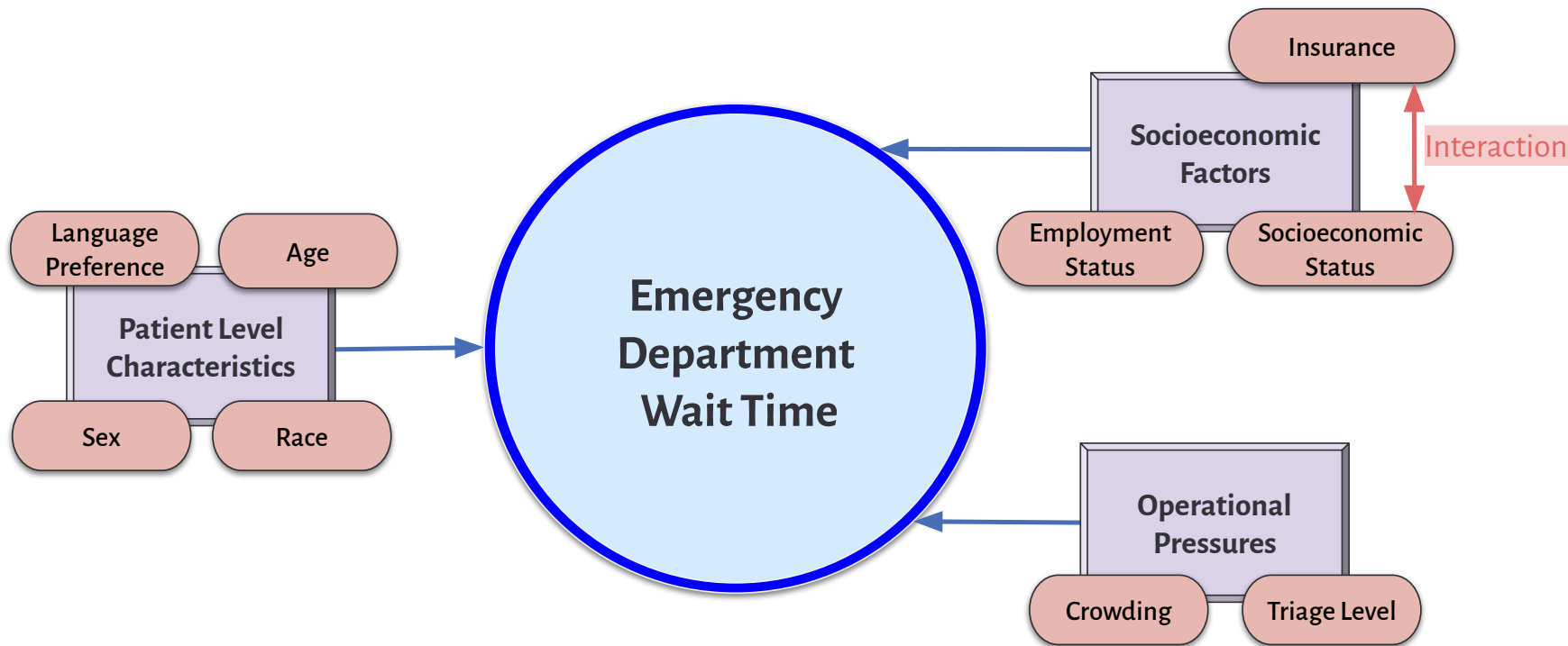
How do demographic and socioeconomic characteristics relate to emergency department wait times, and do these disparities remain after accounting for operational pressures such as triage urgency and ED crowding?

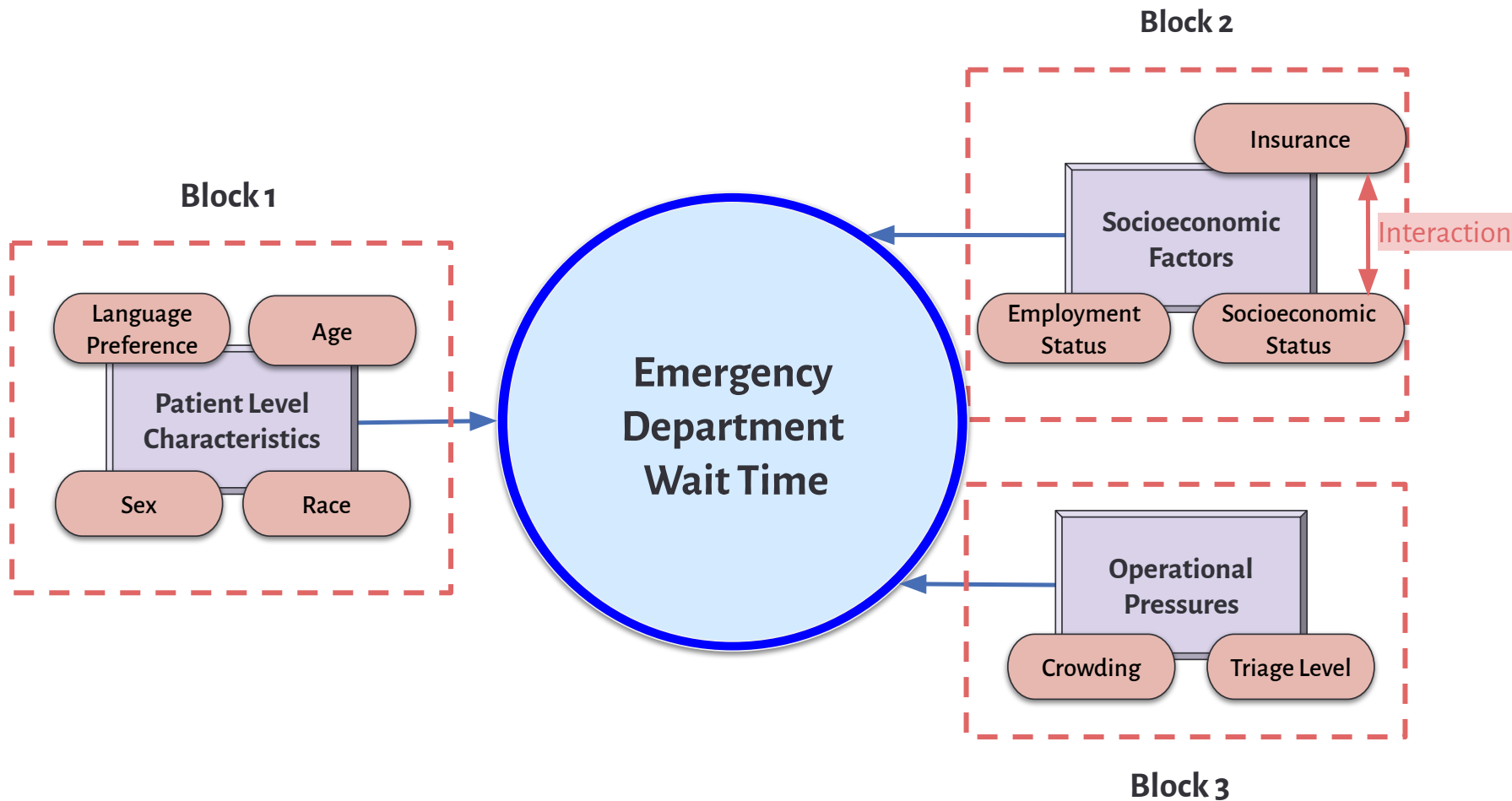
Schematic



Schematic







02.

Dataset



Data + Variables

500

Observations

500 Emergency Department encounters

9

Predictors

Demographic, socioeconomic, and operational variables

1

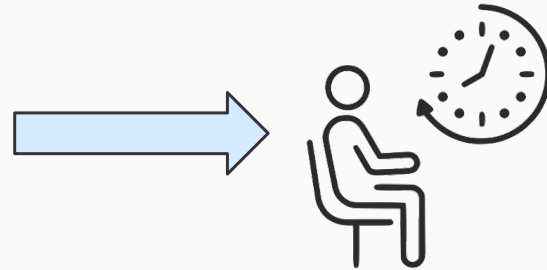
Target Variable

“wait_time_min”

Outcome variable of interest

wait_time_min

number of minutes from ED arrival to being seen by a provider



Data + Variables

Predictors

age_years	Patient's age in years
arrival_hour_oto23	Hour of arrival at the ED (0 = midnight, 23 = 11pm)
day_type	Day of ED visit (e.g., weekday or weekend)
ed_crowding_index	A numeric score indicating ED crowding at the time of arrival
employment_status	Patient's work status (e.g., full-time, part-time, unemployed)
insurance	Type of health insurance held by patient (e.g., private, Medicaid, uninsured)
language_preference	Patient's preferred language for communication (e.g., English, Non-English)

marital_status	Patient's marital status (e.g., single, married, divorced, widowed)
needs_CT	Indicates whether a CT scan was required during the visit
prior_ED_visits_past_year	Number of times patient visited an ED in the past year
race_ethnicity	Patient's racial or ethnic group (e.g., White, Black, Latino)
sex	Patient's sex (male or female)
socioeconomic_status	Patient's combined economic and social position (low, middle, high)
triage_level_1to5	Triage urgency (1 = most urgent, 5 = least urgent)

Data + Variables

Predictors

age_years	Patient's age in years
ed_crowding_index	A numeric score indicating ED crowding at the time of arrival
employment_status	Patient's work status (e.g., full-time, part-time, unemployed)
insurance	Type of health insurance held by patient (e.g., private, Medicaid, uninsured)
language_preference	Patient's preferred language for communication (e.g., English, Non-English)

race_ethnicity	Patient's racial or ethnic group (e.g., White, Black, Latino)
sex	Patient's sex (male or female)
socioeconomic_status	Patient's combined economic and social position (low, middle, high)
triage_level_1to5	Triage urgency (1 = most urgent, 5 = least urgent) (takes into account the severity of injury)

03.

Data Analysis



Methods



EDA

+



Random
Forest

+

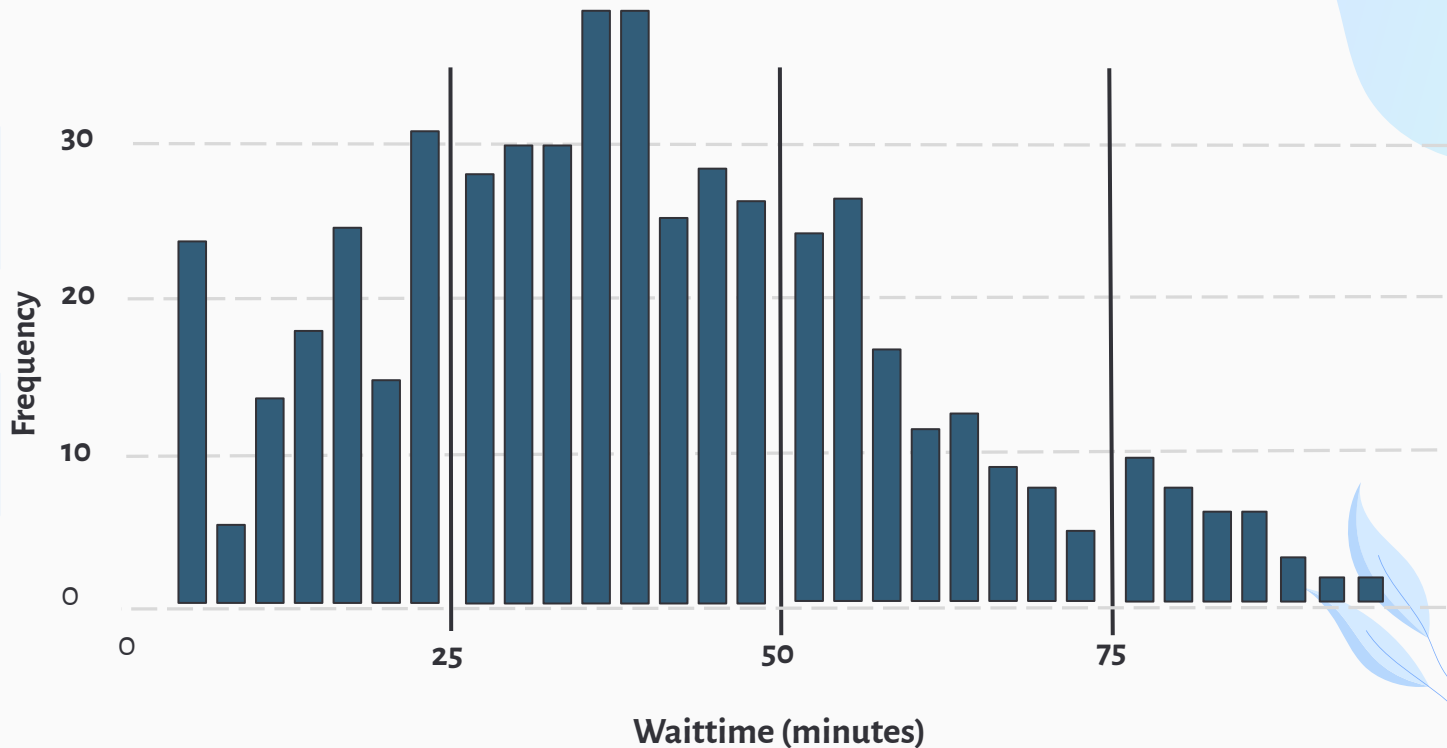


Multiple Linear
Regression

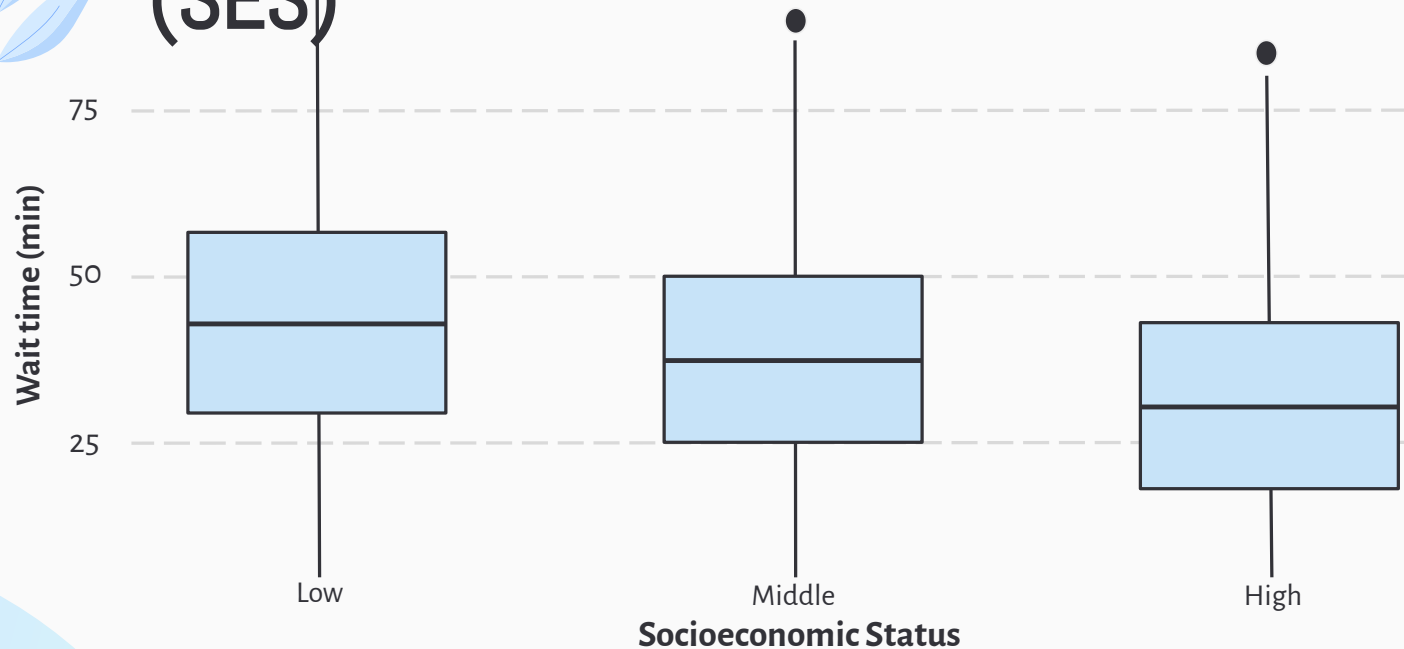
Distribution of ED Wait Times

Most patients wait
between **20-50 mins**

The distribution is
slightly **right-skewed**



ED Wait Time by Socioeconomic Status (SES)

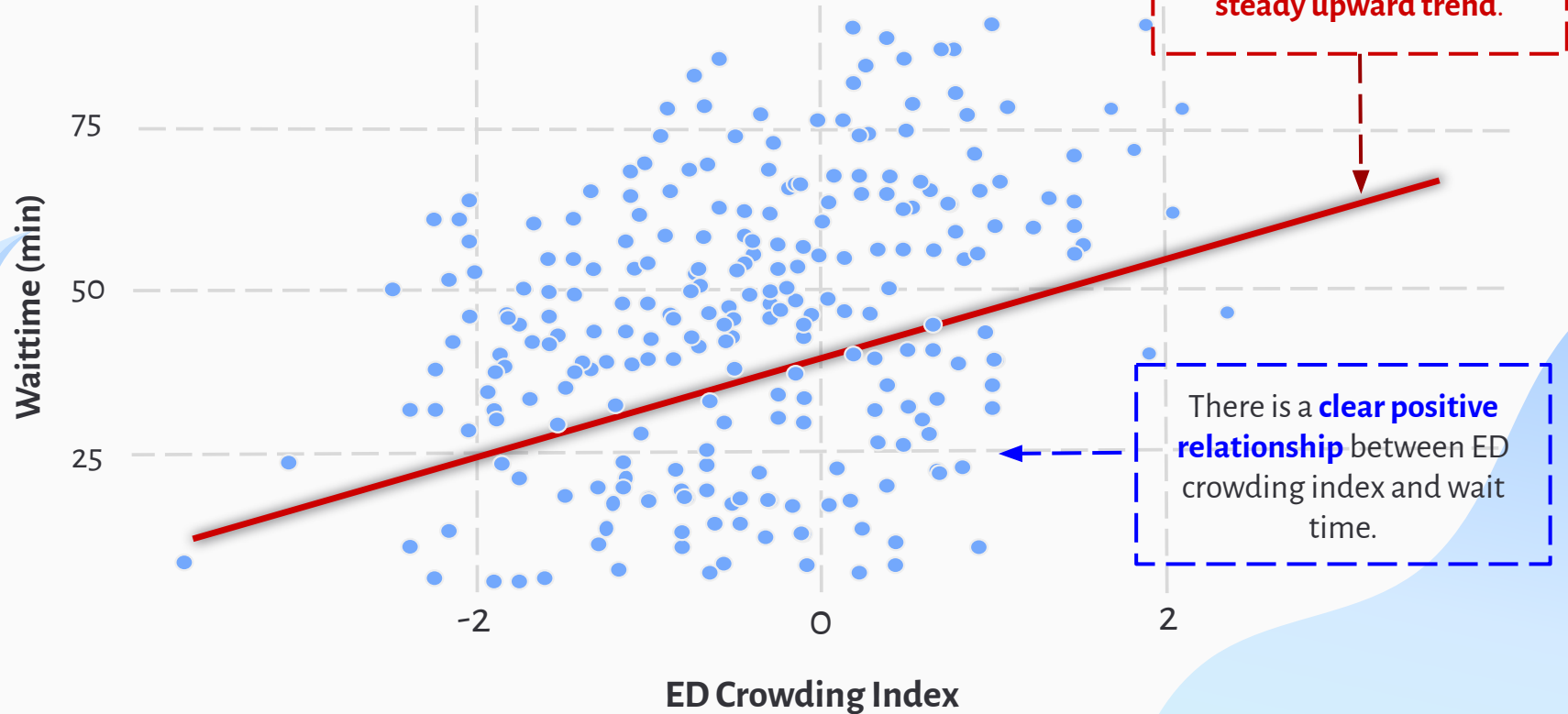


There is a **clear socioeconomic gradient** associated with ED wait times.

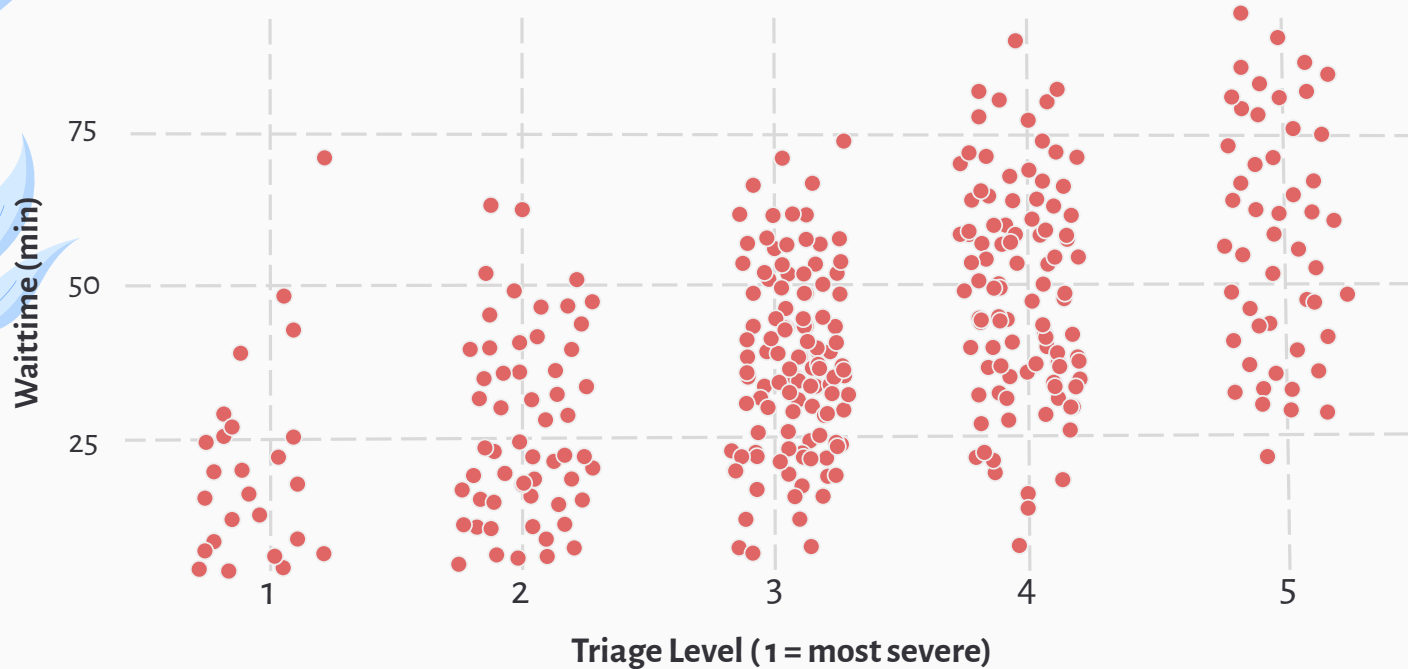
Patients in the **low SES** group show the **longest median wait times**.

High SES patients have the **shortest median wait times**.

ED Crowding and Wait Time



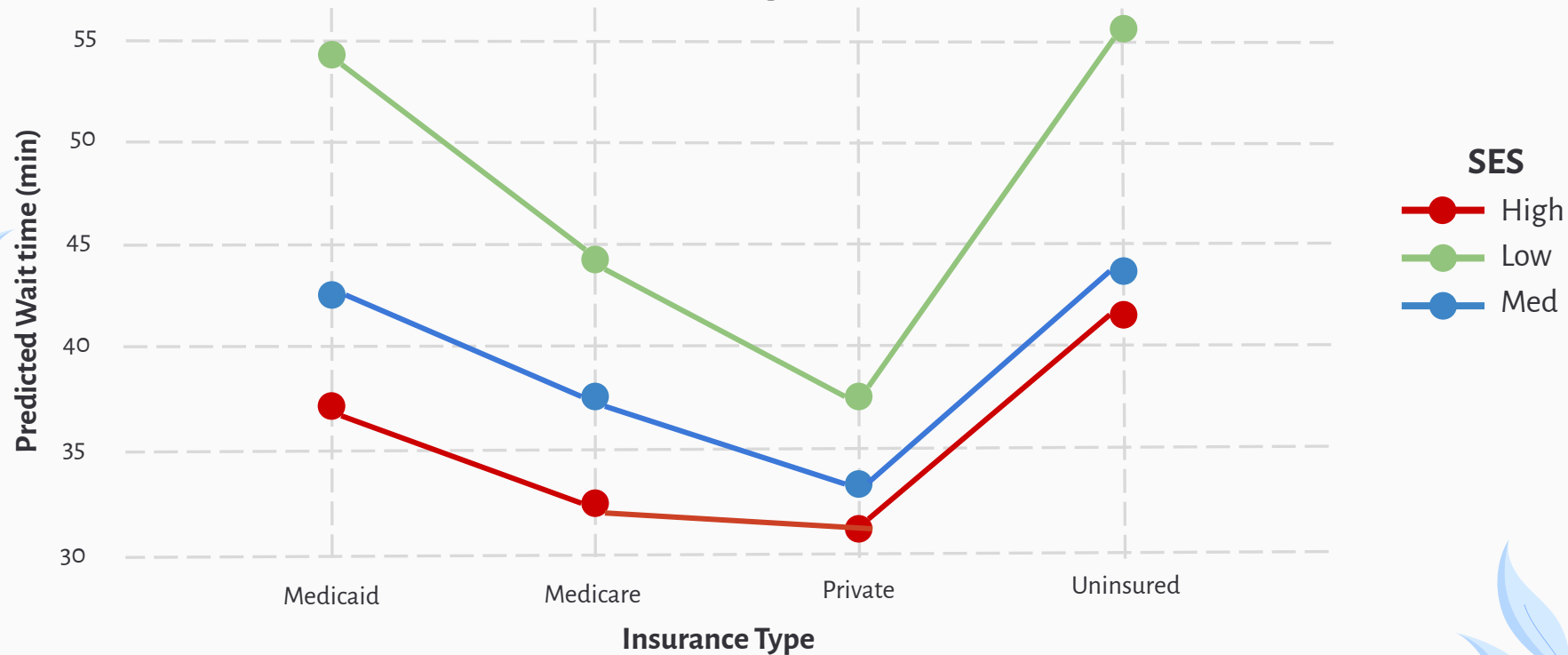
Triage Level vs Wait Time



Higher triage levels
show **longer wait**
times.

Wait time
variability
increases at higher
triage levels.

Insurance x SES Interaction Effect on Predicted ED Wait Time

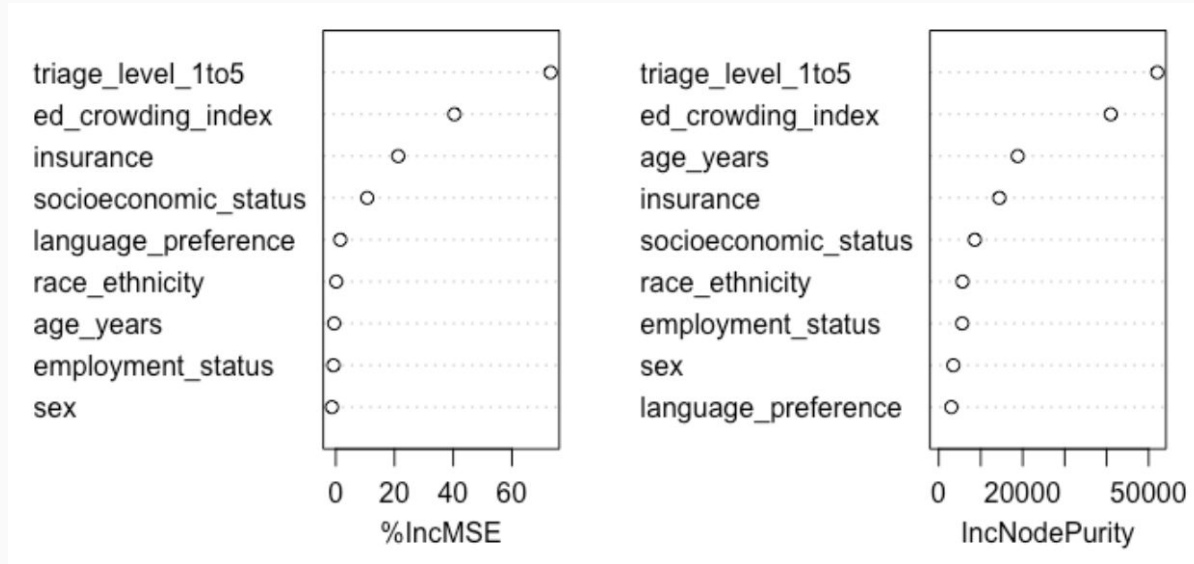


04.

Random Forest



Random Forest Variable Importance for ED Wait Time



Strongest Predictors

- Triage_level_1to5
- Ed_crowding_index
- Insurance type
- Socioeconomic status

05.

MLR Models



Full Multiple Linear Regression Model

wait_time_min ~ ed_crowding_index + triage_level_1to5 + age_years + sex + socioeconomic_status + insurance + language_preference + day_type + arrival_hour_0to23 + needs_CT + prior_ED_visits_past_year + employment_status + marital_status + race_ethnicity

Predictor	<i>b</i>	95% CI [LL, UL]
(Intercept)	21.24**	[14.27, 28.20]
ed_crowding_index	6.92**	[5.90, 7.94]
triage_level_1to52	6.64*	[1.55, 11.73]
triage_level_1to53	17.24**	[12.69, 21.79]
triage_level_1to54	28.55**	[23.81, 33.30]
triage_level_1to55	38.58**	[33.58, 43.58]
age_years	0.01	[-0.04, 0.07]
sexMale	-2.86**	[-4.89, -0.83]
socioeconomic_statusMiddle	3.78**	[0.91, 6.64]
socioeconomic_statusLow	10.54**	[7.31, 13.77]
insuranceMedicare	-6.67**	[-9.91, -3.43]
insurancePrivate	-11.17**	[-14.02, -8.31]
insuranceUninsured	2.72	[-1.33, 6.77]

language_preferencenon-English	1.23	[-1.36, 3.83]
day_typeWeekend	0.90	[-1.37, 3.16]
arrival_hour_0to23	-0.17*	[-0.32, -0.03]
needs_CT	1.27	[-0.95, 3.49]
prior_ED_visits_past_year	-0.44	[-1.37, 0.49]
employment_statusPart-time	1.21	[-1.27, 3.69]
employment_statusUnemployed	0.56	[-2.29, 3.41]
marital_statusMarried	0.38	[-3.10, 3.87]
marital_statusSingle	-0.76	[-4.19, 2.67]
marital_statusWidowed	-0.22	[-5.43, 4.99]
race_ethnicityLatino	0.64	[-2.22, 3.50]
race_ethnicityWhite	-0.33	[-2.84, 2.17]

MLR Model (Theory + Feature Selection)

wait_time_min ~ triage_level_1to5 + ed_crowding_index + insurance * socioeconomic_status + race_ethnicity + needs_CT + arrival_hour_0to23 + age_years + prior_ED_visits_past_year

Key predictors identified by Random Forest and theory-driven variables
+ Also included an SES x insurance interaction

(Intercept)	3.81	[-3.10, 10.71]
triage_level_1to5	10.22**	[9.27, 11.18]
ed_crowding_index	6.89**	[5.88, 7.91]
insuranceMedicare	-5.43	[-12.74, 1.88]
insurancePrivate	-5.89*	[-11.72, -0.07]
insuranceUninsured	4.63	[-5.72, 14.98]
socioeconomic_statusLow	17.45**	[11.06, 23.83]
socioeconomic_statusMiddle	5.38	[-0.77, 11.54]
race_ethnicityLatino	0.63	[-2.19, 3.45]
race_ethnicityWhite	-0.32	[-2.79, 2.15]
needs_CT1	1.15	[-1.05, 3.35]
arrival_hour_0to23	-0.16*	[-0.31, -0.02]
age_years	0.03	[-0.02, 0.09]
prior_ED_visits_past_year	-0.62	[-1.54, 0.30]

insuranceMedicare:socioeconomic_statusLow	-4.92	[-13.93, 4.08]
insurancePrivate:socioeconomic_statusLow	-10.89**	[-18.48, -3.29]
insuranceUninsured:socioeconomic_statusLow	-2.44	[-14.68, 9.80]
insuranceMedicare:socioeconomic_statusMiddle	0.86	[-7.82, 9.53]
insurancePrivate:socioeconomic_statusMiddle	-2.80	[-10.07, 4.47]
insuranceUninsured:socioeconomic_statusMiddle	-2.34	[-14.24, 9.56]

$R^2 = 0.6365$

Blockwise Regression



Block 1: Patient Characteristics

Predictor	<i>b</i>	<i>b</i> 95% CI [LL, UL]
(Intercept)	39.65**	[34.10, 45.21]
age_years	0.08	[-0.00, 0.17]
sexMale	-1.99	[-5.21, 1.23]
race_ethnicityLatino	-4.24	[-8.70, 0.22]
race_ethnicityWhite	-4.46*	[-8.39, -0.53]
language_preferenceNon-English	-1.24	[-5.31, 2.84]

Reference group: Female, Black,
English- preferred patients

$$R^2 = 0.023$$

Patient characteristics do not
account for much of the variance
in ED wait time, only about 2.3%

Block 2: Socioeconomic Factors Included

Predictor	<i>b</i>	<i>b</i> 95% CI [LL, UL]
(Intercept)	39.66**	[32.91, 46.42]
age_years	0.06	[-0.02, 0.14]
sexMale	-2.81	[-5.81, 0.18]
race_ethnicityLatino	-1.67	[-5.83, 2.48]
race_ethnicityWhite	-2.92	[-6.57, 0.73]
language_preferencenon-English	0.24	[-3.57, 4.06]
insuranceMedicare	-8.13**	[-12.85, -3.41]
insurancePrivate	-12.46**	[-16.66, -8.26]
insuranceUninsured	2.65	[-3.30, 8.59]

socioeconomic_statusLow	13.22**	[8.47, 17.97]
socioeconomic_statusMiddle	6.60**	[2.39, 10.81]
employment_statusPart-time	1.12	[-2.50, 4.75]
employment_statusUnemployed	-2.30	[-6.48, 1.89]

Reference group: Female, Black, English-preferred, Medicaid insurance, High SES, Full-time employment patients

$$R^2 = 0.179$$



$$\Delta R^2 = 0.156$$

Socioeconomic factors explains about 15.6% additional variance

Block 3: Operational Pressures Included

Predictor	<i>b</i>	<i>b</i> 95% CI [LL, UL]
(Intercept)	6.10*	[0.59, 11.62]
age_years	0.02	[-0.04, 0.07]
sexMale	-2.94**	[-4.96, -0.93]
race_ethnicityLa tino	0.43	[-2.39, 3.25]
race_ethnicityW hite	-0.60	[-3.08, 1.87]
language_prefer enceNon- English	1.29	[-1.29, 3.86]
insuranceMedic are	-6.53**	[-9.73, -3.33]
insurancePrivate	-11.30**	[-14.14, -8.47]
insuranceUninsu red	2.20	[-1.80, 6.21]
socioeconomic_ statusLow	10.28**	[7.06, 13.49]
socioeconomic_ statusMiddle	3.82**	[0.97, 6.68]
employment_sta tusPart-time	1.66	[-0.79, 4.11]
employment_sta tusUnemployed	0.71	[-2.12, 3.54]

ed_crowding_in dex	6.85**	[5.84, 7.87]
triage_level_1to 5	10.27**	[9.31, 11.23]

$$R^2 = 0.628$$



$$\Delta R^2 = 0.449$$

Operational Pressure variables explain about 44.9% additional variance, far more than patient or socioeconomic factors.

VIF values: ranged from ~1.01 to ~1.073
→ no multicollinearity issues

Block 3: Operational Pressures Included

Predictor	<i>b</i>	<i>b</i> 95% CI [LL, UL]
(Intercept)	19.31**	[13.45, 25.16]
triage_level_1to52	6.26*	[1.21, 11.32]
triage_level_1to53	16.95**	[12.42, 21.49]
triage_level_1to54	28.30**	[23.58, 33.02]
triage_level_1to55	38.31**	[33.34, 43.28]
ed_crowding_index	6.86**	[5.85, 7.87]
age_years	0.02	[-0.03, 0.07]
sexMale	-2.98**	[-5.00, -0.97]
language_preferenceNon-English	1.35	[-1.23, 3.93]
insuranceMedicare	-6.42**	[-9.63, -3.20]
insurancePrivate	-11.24**	[-14.07, -8.41]
insuranceUninsured	2.34	[-1.66, 6.35]
socioeconomic_statusLow	10.18**	[6.98, 13.38]

socioeconomic_statusMiddle	3.64*	[0.79, 6.50]
employment_statusPart-time	1.52	[-0.94, 3.97]
employment_statusUnemployed	0.56	[-2.27, 3.39]

$$R^2 = 0.630$$



$$\Delta R^2 = 0.449$$

Operational Pressure variables explain about 44.9% additional variance, far more than patient or socioeconomic factors.

VIF values: ranged from ~1.01 to ~1.073
→ no multicollinearity issues

Model with Interaction

Predictor	<i>b</i>	<i>b</i> 95% CI [LL, UL]
(Intercept)	15.89**	[8.98, 22.79]
triage_level_1to52	6.15*	[1.14, 11.17]
triage_level_1to53	16.93**	[12.43, 21.43]
triage_level_1to54	28.23**	[23.54, 32.91]
triage_level_1to55	38.21**	[33.28, 43.15]
ed_crowding_index	6.79**	[5.79, 7.80]
age_years	0.03	[-0.02, 0.08]
sexMale	-3.14**	[-5.16, -1.12]
language_prefer enceNon- English	1.49	[-1.08, 4.06]
insuranceMedicare	-4.29	[-11.58, 3.00]
insurancePrivate	-6.17*	[-11.95, -0.39]
insuranceUninsured	3.83	[-6.53, 14.20]
socioeconomic_statusLow	17.91**	[11.34, 24.48]

socioeconomic_statusMiddle	5.04	[-1.13, 11.21]
employment_statusPart-time	1.23	[-1.22, 3.67]
employment_statusUnemployed	0.59	[-2.22, 3.40]
insuranceMedicare:socioeconomic_statusLow	-6.57	[-15.53, 2.40]
insurancePrivate:socioeconomic_statusLow	-11.90**	[-19.40, -4.39]
insuranceUninsured:socioeconomic_statusLow	-3.21	[-15.44, 9.01]
insuranceMedicare:socioeconomic_statusMiddle	-0.25	[-8.90, 8.39]
insurancePrivate:socioeconomic_statusMiddle	-2.32	[-9.52, 4.89]
insuranceUninsured:socioeconomic_statusMiddle	-1.63	[-13.58, 10.31]

SES x Insurance
interaction

$R^2 = 0.641$

Low-SES private patients wait ~12 mins longer than high-SES private patients, even after controlling for triage and ED crowding

BIC Stepwise Model

Predictor	<i>b</i>	<i>b</i> 95% CI [LL, UL]
(Intercept)	5.95**	[1.63, 10.27]
triage_level_1to5	10.26**	[9.31, 11.21]
ed_crowding_in dex	6.91**	[5.89, 7.92]
insuranceMedicare	-6.16**	[-9.34, -2.98]
insurancePrivate	-11.01**	[-13.84, -8.17]
insuranceUninsured	2.43	[-1.59, 6.44]
socioeconomic_statusLow	10.43**	[7.55, 13.30]
socioeconomic_statusMiddle	3.80**	[1.01, 6.58]

BIC model: applies a stronger penalty for model complexity, drops weaker predictors, resulting model is more concise

Dropped:
needs_CT, race_ethnicity, age_years,
needs_CT, arrival_hour_oto23,
prior_ED_visits_past_year

BIC Stepwise Model

Predictor	<i>b</i>	<i>b</i> 95% CI [LL, UL]
(Intercept)	5.95**	[1.63, 10.27]
triage_level_1to5	10.26**	[9.31, 11.21]
ed_crowding_index	6.91**	[5.89, 7.92]
insuranceMedicare	-6.16**	[-9.34, -2.98]
insurancePrivate	-11.01**	[-13.84, -8.17]
insuranceUninsured	2.43	[-1.59, 6.44]
socioeconomic_statusLow	10.43**	[7.55, 13.30]
socioeconomic_statusMiddle	3.80**	[1.01, 6.58]

VIF Values

	GVIF	Df	GVIF ^{1/(2*Df)}
triage_level_1to5	1.018827	1	1.009369
ed_crowding_index	1.020311	1	1.010104
insurance	1.029287	3	1.004823
socioeconomic_status	1.024698	2	1.006118

The VIF values range from ~1.0 to 1.01
→ indicates no concerns of multicollinearity

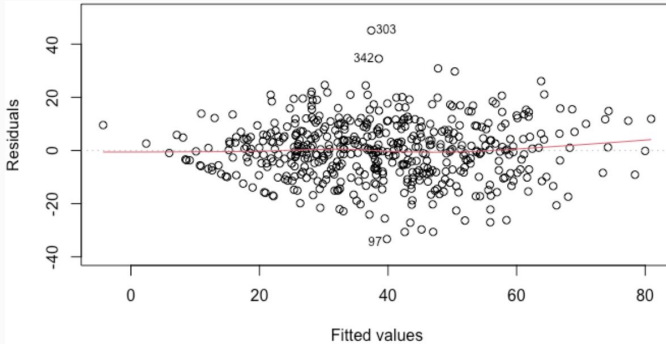
06.

Model Diagnostics



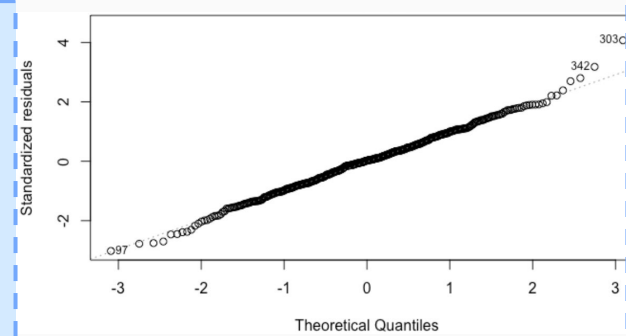
Full Model (Block 3 Model) Diagnostics

Residuals vs Fitted



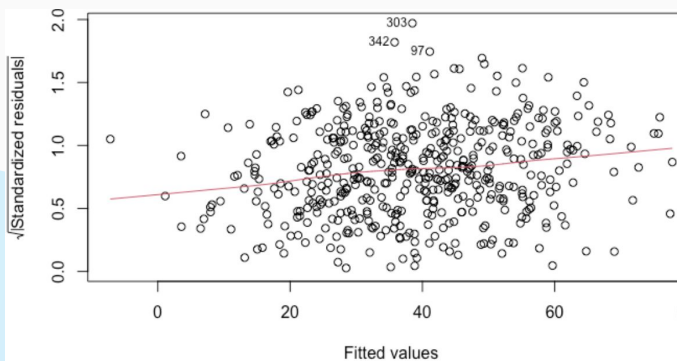
Residuals are generally randomly scattered $\rightarrow$ no major violation of homoscedasticity

Normal Q-Q



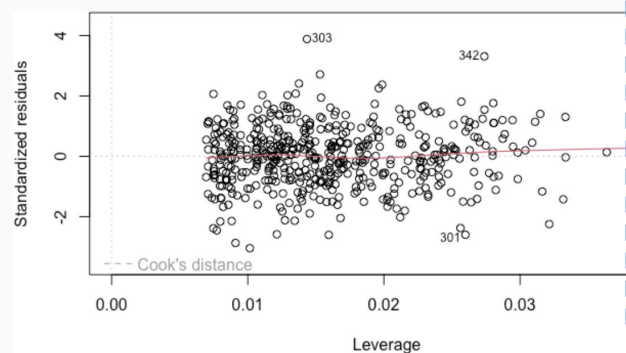
Points generally fall along the diagonal reference line $\rightarrow$ residuals approx. normally distributed

Scale-Location



Fairly constant variance across fitted values $\rightarrow$ confirms homoscedasticity

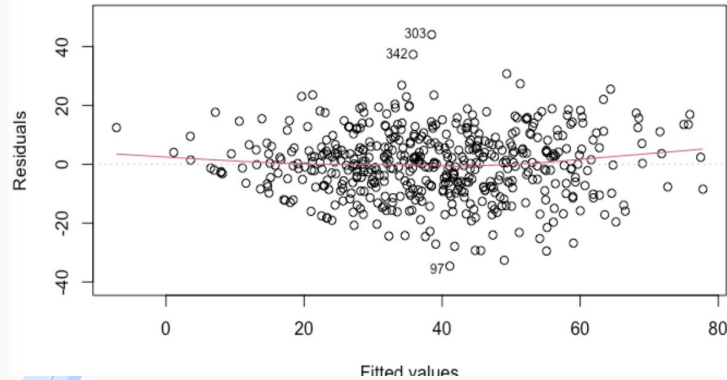
Residuals vs Leverage



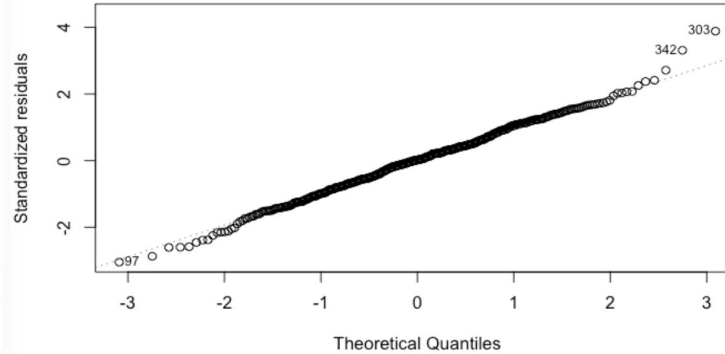
No influential outliers or high-leverage problems

BIC Model Diagnostics

Residuals vs Fitted



Normal Q-Q



Similar to full model diagnostics:

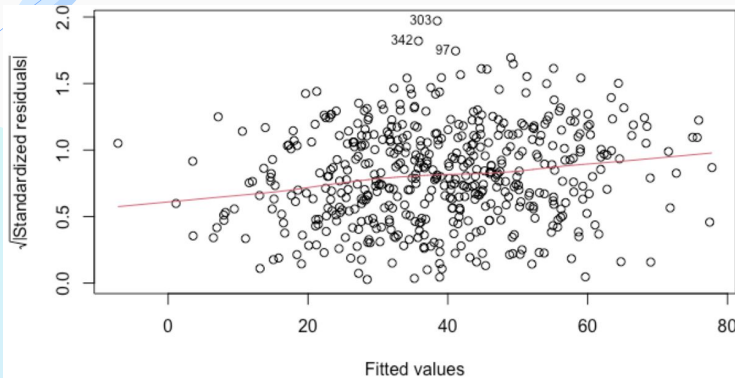
Homoscedasticity assumption met

Residuals approx. normally distributed

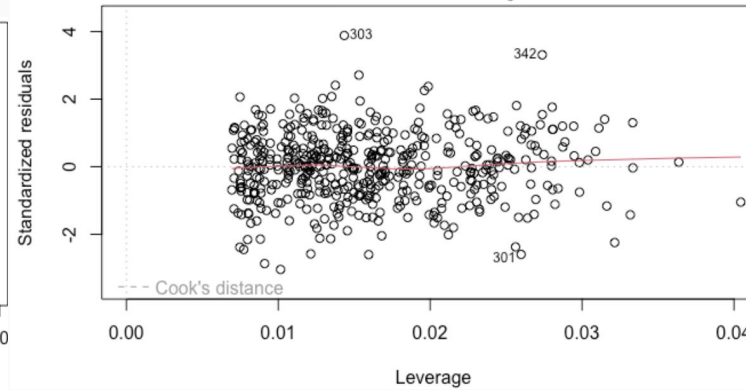
No influential outliers or high leverage problems

→ variable reduction process did not sacrifice model fit or assumption validity

Scale-Location



Residuals vs Leverage



Results



Demographic Characteristics and ED Wait Time

- Basic demographics explain very little of wait time ($\approx 2\%$)
- Age, race, sex, and language show minimal predictive power compared to other factors



Operational Pressures and ED Wait Time

- Triage urgency and ED crowding explain the largest share of variation ($\approx 45\%$)
- These are the dominant drivers of wait time across all patients



Socioeconomic Factors and ED Wait Time

- SES and insurance meaningfully increase explained variance ($\approx 15\%$).
- Low SES and Medicaid are associated with longer wait times, while private insurance reduces wait time



SES $\times$ Insurance Interaction

- Insurance benefits depend on SES
- Low-SES private insurance holders wait much longer than high-SES private patients

Conclusion

Significant Predictors



Triage Level



ED Crowding



Insurance



SES

Back to our Research Question:

How do demographic and socioeconomic characteristics relate to emergency department wait times, and do these disparities remain after accounting for operational pressures such as triage urgency and ED crowding?

→ Demographic traits explain very little of ED wait time, while socioeconomic status and insurance type more largely and meaningfully impact how long patients wait

→ After accounting for triage urgency and ED crowding, socioeconomic disparities still persist:

- low-SES patients with private insurance still wait significantly longer than high-SES private patients

Thus, operational pressures are the largest predictors overall, but socioeconomic inequities remain even in similarly crowded ED conditions

Limitations + Suggestions

Synthetic dataset (not real patient data)

Patterns may not reflect true ED variability / perfectly generalize to real ED operations



Validate findings using real ED operational data

Interaction model captures only SES x Insurance



Consider and test additional interactions: race x SES, race x insurance, three-way interactions

Some literature referenced comes from healthcare systems outside the U.S. which may differ in insurance structures and ED operations



Compare with more U.S.-based ED studies, explore cross-system disparities

S

James, C. A., Bourgeois, F. T., & Shannon, M. W. (2005). Association of race/ethnicity with emergency department wait times. *Pediatrics*, 115(3), e310–e315.
<https://doi.org/10.1542/peds.2004-1541>

Turner, A. J., Francetic, I., Watkinson, R., Gillibrand, S., & Sutton, M. (2022). Socioeconomic inequality in access to timely and appropriate care in emergency departments. *Journal of Health Economics*, 85, 102668. <https://doi.org/10.1016/j.jhealeco.2022.102668>

Thank you for
listening!

